

Analytical Gradient Approximation of Affirmation Loss for Efficient Token Attribution*

Arav Santhanam

May 22, 2026

1 Method Overview and Affirmation Loss

The Token Highlighter ([see original paper](#)) inspects and mitigates adversarial jailbreak threats in user queries by computing per-token influence scores using the ℓ_2 -norm of the input gradients, $\left\| \frac{\partial \mathcal{L}}{\partial \mathbf{x}_i} \right\|_2$. The **Affirmation Loss**, $\mathcal{L}$, evaluates the negative log-likelihood of a predefined target phrase $\mathbf{y}$ conditioned on a user query with token embeddings $\mathbf{x}_i$. The method applies soft removal by shrinking the embeddings of ‘driver tokens’ encouraging model compliance.

In production serving systems such as vLLM, executing a standard full autograd backward pass during inference introduces prohibitive latency and memory overhead, as intermediate activations are discarded to optimize paged memory layouts. Our analytical formulation avoids this bottleneck. It requires only a single forward pass through the prefill stage to extract attention activations and weight matrices via hooks, mapping token attributions back along the sequence axis without the need for a global backpropagation graph.

Index the concatenated prompt and response tokens as the set $R = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n, \mathbf{j}_0, \mathbf{j}_1, \dots, \mathbf{j}_m\}$, where prompt tokens are 1-indexed and generated target tokens are 0-indexed. Define $t = n + j$ as the absolute sequence position of the j -th generated token, where the target generation window spans $t \in T = \{n, n + 1, \dots, n + m - 1\}$. Let $\mathbf{z}_t = W_U \mathbf{h}_t^{(L)}$ be the logit vector at step t , where $W_U \in \mathbb{R}^{V \times d_{\text{model}}}$ is the unembedding matrix. The affirmation loss is the teacher-forced negative log-likelihood over the target affirmative sequence $\mathbf{y}$:

$$\mathcal{L}(\mathbf{x}_{1:n}) = - \sum_{t \in T} \log P_{\theta}(y_{t-n} \mid \mathbf{x}_{1:n}, y_{<t-n}) = - \sum_{t \in T} \log \sigma(\mathbf{z}_t)_{y_{t-n}}, \quad (1)$$

where $\sigma(\cdot)$ is the softmax operator and $\sigma(\mathbf{z}_t)_{y_{t-n}}$ is the scalar probability assigned to the ground-truth affirmative target token at step t .

2 Residual Stream Approximation

In an L -layer decoder-only transformer, the input embedding $\mathbf{x}_i$ initializes the residual stream at sequence position i , which accumulates downstream layer transformations:

$$\mathbf{h}_i^{(L-1)} = \mathbf{x}_i + \sum_{\ell=1}^{L-1} (\text{Attn}_i^{(\ell)} + \text{FFN}_i^{(\ell)}),$$

*This work is part of an ongoing collaboration with Pin-Yu Chen and Irene Ko at IBM Research.

where $\mathbf{h}_i^{(L-1)}$ is the hidden state vector at position i following layer $L - 1$. Because $\mathbf{x}_i$ propagates through successive skip-connections, the analytical Jacobian $\frac{\partial \mathbf{h}_i^{(L-1)}}{\partial \mathbf{x}_i}$ expands to the identity matrix I plus a matrix of cross-token feedback tracking downstream attention dependencies.

Assuming that direct identity paths dominate local sensitivity, we omit the cross-token feedback components and approximate the transformation boundary as:

$$\frac{\partial \mathbf{h}_i^{(L-1)}}{\partial \mathbf{x}_i} \approx I \implies \frac{\partial \mathcal{L}}{\partial \mathbf{x}_i} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} \cdot \frac{\partial \mathbf{h}_i^{(L-1)}}{\partial \mathbf{x}_i} \approx \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}}.$$

Evaluating the total derivative via the multivariate Chain Rule along the sequence axis yields the summation:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} = \sum_{t \in T} \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \frac{\partial \mathbf{h}_t^{(L)}}{\partial \mathbf{h}_i^{(L-1)}}. \quad (2)$$

For a fixed target generation step $t \in T$, we derive the boundary gradient $\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}}$ and the token projection Jacobian $\frac{\partial \mathbf{h}_t^{(L)}}{\partial \mathbf{h}_i^{(L-1)}}$ independently.

Remark 1 (Architectural Scope and Approximations). *While the conceptual formulation of Affirmation Loss gradients applies universally to any conditional sequence pipeline, this derivation explicitly assumes a standard decoder-only autoregressive transformer layout. This assumption provides three operational properties:*

- (i) *Prompt tokens and generated text share a unified sequence axis, allowing cross-token dependencies to be mapped via a single causal self-attention matrix.*
- (ii) *The language modeling head is a linear projection W_U , enabling the exact analytical isolation of the boundary gradient.*
- (iii) *The underlying software runtime (e.g. vLLM) supports the extraction of attention activations $(\alpha, \mathbf{q}, \mathbf{k}, \mathbf{v})$ via forward-pass hooks.*

Finally, as formalized in Section 2, this framework omits secondary cross-token feedback from intermediate layers, isolating the local input sensitivity along the direct identity path of the residual stream.

3 Gradient of the Language Modeling (LM) Head

We isolate the loss gradient with respect to the final hidden state at step t , denoted $\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}}$. Let s index positions across the full concatenated prompt-response sequence. Expanding via the multivariate Chain Rule yields:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} = \sum_{s=0}^{n+m-1} \frac{\partial \mathcal{L}}{\partial \mathbf{z}_s} \frac{\partial \mathbf{z}_s}{\partial \mathbf{h}_t^{(L)}} = \underbrace{\sum_{s=0}^{n-1} \frac{\partial \mathcal{L}}{\partial \mathbf{z}_s} \frac{\partial \mathbf{z}_s}{\partial \mathbf{h}_t^{(L)}}}_{\text{Prompt Tokens (0)}} + \underbrace{\sum_{s=n}^{n+m-1} \frac{\partial \mathcal{L}}{\partial \mathbf{z}_s} \frac{\partial \mathbf{z}_s}{\partial \mathbf{h}_t^{(L)}}}_{\text{Generated Tokens}} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}_t} \frac{\partial \mathbf{z}_t}{\partial \mathbf{h}_t^{(L)}}. \quad (3)$$

The prompt-token summation vanishes because the support of the affirmation loss does not extend to logits evaluated during the prefill phase ($\frac{\partial \mathcal{L}}{\partial \mathbf{z}_s} = \mathbf{0}$ for $s < n$). Furthermore, since

the logit vector $\mathbf{z}_s$ depends strictly on the concurrent local hidden state ($\mathbf{z}_t = W_U \mathbf{h}_t^{(L)}$), the cross-position derivative $\frac{\partial \mathbf{z}_s}{\partial \mathbf{h}_t^{(L)}} = 0$ for all $s \neq t$.

To find $\frac{\partial \mathcal{L}}{\partial \mathbf{z}_t}$, we differentiate the scalar loss contribution at step t , $\mathcal{L}_t = -\log \sigma(\mathbf{z}_t)_{y_{t-n}}$, with respect to an arbitrary logit element $z_{t,k}$:

$$\frac{\partial \mathcal{L}_t}{\partial z_{t,k}} = -\frac{\partial}{\partial z_{t,k}} (\log \sigma(\mathbf{z}_t)_{y_{t-n}}) = -\frac{1}{\sigma(\mathbf{z}_t)_{y_{t-n}}} \cdot \frac{\partial \sigma(\mathbf{z}_t)_{y_{t-n}}}{\partial z_{t,k}}.$$

The Jacobian of the softmax function is given by:

$$\frac{\partial \sigma(\mathbf{z})_k}{\partial z_r} = \sigma(z_k)(\delta_{kr} - \sigma(z_r)),$$

where δ_{kr} is the Kronecker delta. Substituting this identity into the coordinate derivative yields:

$$\frac{\partial \mathcal{L}_t}{\partial z_{t,k}} = -\frac{1}{\sigma(\mathbf{z}_t)_{y_{t-n}}} \cdot [\sigma(\mathbf{z}_t)_{y_{t-n}} (\delta_{y_{t-n},k} - \sigma(\mathbf{z}_t)_k)].$$

Canceling the scalar probability mass term $\sigma(\mathbf{z}_t)_{y_{t-n}}$ simplifies the expression to:

$$\frac{\partial \mathcal{L}_t}{\partial z_{t,k}} = -(\delta_{y_{t-n},k} - \sigma(\mathbf{z}_t)_k) = \sigma(\mathbf{z}_t)_k - \delta_{y_{t-n},k} = p_{t,k} - \delta_{y_{t-n},k}.$$

Letting $\mathbf{p}_t = \sigma(\mathbf{z}_t) \in \mathbb{R}^V$ denote the dense vocabulary probability vector and collecting these coordinate derivatives into a vector yields:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{z}_t} = \mathbf{p}_t - \mathbf{e}_{y_{t-n}}, \quad (4)$$

where $\mathbf{e}_{y_{t-n}} \in \mathbb{R}^V$ is the standard one-hot basis vector at the ground-truth target token index.

Given the linear formulation of the language modeling head, $\mathbf{z}_t = W_U \mathbf{h}_t^{(L)}$, the sensitivity matrix is $\frac{\partial \mathbf{z}_t}{\partial \mathbf{h}_t^{(L)}} = W_U$. Substituting these terms back into Equation (3) via the vector chain rule determines the exact boundary condition for the final hidden states:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} = W_U^\top (\mathbf{p}_t - \mathbf{e}_{y_{t-n}}). \quad (5)$$

The vector $\left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right) \in \mathbb{R}^{d_{\text{model}}}$ acts as the final layer error signal at absolute sequence position t .

4 Gradient via Value and Key Paths

4.1 Back-projecting Gradients Through the Final Attention Layer

We expand the remaining factor from Equation (2), tracking the token projection Jacobian $\frac{\partial \mathbf{h}_t^{(L)}}{\partial \mathbf{h}_i^{(L-1)}}$ backward through the final causal attention block. This decomposition balances two distinct channels originating from prompt token $\mathbf{x}_i$:

1. **Value Path:** Tracks semantic magnitude, mapping how updates in the representation of prompt token $\mathbf{x}_i$ propagate features directly into the residual stream via the value vector transformation.

2. **Key Path:** Tracks routing variations, measuring how changes in the key vector of prompt token $\mathbf{x}_i$ modulate the causal attention weights.

Within the final attention block, the hidden state vector at position t expands as:

$$\mathbf{h}_t^{(L)} = \mathbf{h}_t^{(L-1)} + W_O \sum_{k=0}^t \alpha_{tk} \mathbf{v}_k,$$

where $W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ is the attention output projection matrix. Both the attention weight α_{ti} and the value vector $\mathbf{v}_i = W_V \mathbf{h}_i^{(L-1)}$ depend explicitly on the prompt hidden state $\mathbf{h}_i^{(L-1)}$ (via the key projection $\mathbf{k}_i = W_K \mathbf{h}_i^{(L-1)}$ and linear transformation, respectively). Applying the product rule to the prompt index column $k = i$ isolates the total Jacobian of the raw attention context vector $\mathbf{o}_t = \sum_k \alpha_{tk} \mathbf{v}_k$:

$$\frac{\partial \mathbf{o}_t}{\partial \mathbf{h}_i^{(L-1)}} = \underbrace{\alpha_{ti} W_V}_{\text{Value Path}} + \underbrace{\sum_{k=0}^t \mathbf{v}_k \left(\frac{\partial \alpha_{tk}}{\partial \mathbf{h}_i^{(L-1)}} \right)^\top}_{\text{Key Path}},$$

which yields the layer hidden state projection Jacobian:

$$\frac{\partial \mathbf{h}_t^{(L)}}{\partial \mathbf{h}_i^{(L-1)}} = W_O \left[\alpha_{ti} W_V + \sum_{k=0}^t \mathbf{v}_k \left(\frac{\partial \alpha_{tk}}{\partial \mathbf{h}_i^{(L-1)}} \right)^\top \right].$$

4.2 Value Path Contribution

Isolating the value path by holding the attention distribution weights constant, the structural mapping reduces to the linear operator $W_O \alpha_{ti} W_V$. Projecting the error gradient vector $\left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right)$ from the vocabulary layer backward and accumulating across the generation window yields:

$$\left. \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} \right|_{\text{value}} = \sum_{t \in T} (W_O \alpha_{ti} W_V)^\top \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} = W_V^\top W_O^\top \sum_{t \in T} \alpha_{ti} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right).$$

5 Attention Mechanism Gradients

Let $\mathbf{q}_t$, $\mathbf{k}_i$, and $\mathbf{v}_i$ denote the query, key, and value vectors at positions t and i . The pre-softmax alignment score s_{ti} and normalized attention weight α_{ti} are defined as:

$$s_{ti} = \frac{\mathbf{q}_t^\top \mathbf{k}_i}{\sqrt{d}}, \quad \alpha_{ti} = \sigma(s_{ti}),$$

where d is the head dimension. Differentiating the attention score α_{tk} with respect to the raw scalar alignment input s_{ti} yields:

$$\frac{\partial \alpha_{tk}}{\partial s_{ti}} = \alpha_{tk} (\delta_{ki} - \alpha_{ti}).$$

5.1 Attention Output Gradient w.r.t. Pre-Softmax Scores

Expanding the total derivative of the context vector $\mathbf{o}_t$ relative to the raw score s_{ti} yields:

$$\begin{aligned}\frac{\partial \mathbf{o}_t}{\partial s_{ti}} &= \sum_{k=0}^t \frac{\partial \alpha_{tk}}{\partial s_{ti}} \mathbf{v}_k = \sum_{k=0}^t \alpha_{tk} (\delta_{ki} - \alpha_{ti}) \mathbf{v}_k \\ &= \sum_{k=0}^t \alpha_{tk} \delta_{ki} \mathbf{v}_k - \alpha_{ti} \sum_{k=0}^t \alpha_{tk} \mathbf{v}_k \\ &= \alpha_{ti} \mathbf{v}_i - \alpha_{ti} \bar{\mathbf{o}}_t = \alpha_{ti} (\mathbf{v}_i - \bar{\mathbf{o}}_t),\end{aligned}$$

where $\bar{\mathbf{o}}_t = \sum_k \alpha_{tk} \mathbf{v}_k \in \mathbb{R}^{d_{\text{head}}}$ is the expected attention-weighted mean of the value vectors at position t . This identity shows that an increase in s_{ti} drives the context vector toward $\mathbf{v}_i$ while suppressing competing components proportional to the current expectation.

5.2 Alignment Scalar and Softmax Baseline

We map the sensitivity of the loss function with respect to the pre-softmax coordinate s_{ti} using the contextual linear chains:

$$\frac{\partial \mathcal{L}}{\partial s_{ti}} = \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right)^\top \frac{\partial \mathbf{h}_t^{(L)}}{\partial \mathbf{o}_t} \frac{\partial \mathbf{o}_t}{\partial s_{ti}} = \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right)^\top W_O \cdot \alpha_{ti} (\mathbf{v}_i - \bar{\mathbf{o}}_t).$$

Define the scalar alignment score g_{ti} as the inner product mapping how closely an input value vector aligns with the back-projected loss optimization coordinate:

$$g_{ti} := (\mathbf{p}_t - \mathbf{e}_{y_{t-n}})^\top W_U W_O \mathbf{v}_i = \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right)^\top W_O \mathbf{v}_i.$$

Similarly, define the baseline $\bar{g}_t$ as the expected value of token alignments matching the active attention weights:

$$\bar{g}_t := \sum_{k=0}^t \alpha_{tk} g_{tk}.$$

Because modifying an individual pre-softmax score contracts the remaining mass uniformly, the optimization sensitivity reduces to the centered alignment value:

$$\frac{\partial \mathcal{L}}{\partial s_{ti}} = \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right)^\top W_O \alpha_{ti} (\mathbf{v}_i - \bar{\mathbf{o}}_t) = \alpha_{ti} (g_{ti} - \bar{g}_t).$$

5.3 Key Path Contribution

Given the scaling mechanism $s_{ti} = \frac{\mathbf{q}_t^\top W_K \mathbf{h}_i^{(L-1)}}{\sqrt{d}}$, the input parameter gradient is $\frac{\partial s_{ti}}{\partial \mathbf{h}_i^{(L-1)}} = \frac{W_K^\top \mathbf{q}_t}{\sqrt{d}}$. Accumulating this routing vector across the generation span defines the key path metric:

$$\left. \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} \right|_{\text{key}} = \sum_{t \in T} \frac{\partial \mathcal{L}}{\partial s_{ti}} \frac{\partial s_{ti}}{\partial \mathbf{h}_i^{(L-1)}} = \frac{W_K^\top}{\sqrt{d}} \sum_{t \in T} \alpha_{ti} (g_{ti} - \bar{g}_t) \mathbf{q}_t.$$

5.4 Complete Analytical Gradient Formulation

Accumulating the terms derived via the product rule across both paths yields the total analytical gradient evaluating the prompt token representation at position i :

$$\boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} = \underbrace{W_V^\top W_O^\top \sum_{t \in T} \alpha_{ti} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right)}_{\text{Value Path Magnitude}} + \underbrace{\frac{W_K^\top}{\sqrt{d}} \sum_{t \in T} \alpha_{ti} (g_{ti} - \bar{g}_t) \mathbf{q}_t}_{\text{Key Path Routing Sensitivity}}} \quad (6)$$

The **value path** isolates direct semantic contribution: the cumulative back-projected error vector $\sum_{t \in T} \alpha_{ti} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_t^{(L)}} \right)$ maps directly into the input residual dimensions via $W_V^\top W_O^\top$. Conversely, the **key path** captures attention routing updates: the accumulated query context is scaled by the centered sensitivity coefficient $\alpha_{ti}(g_{ti} - \bar{g}_t)$. This component has large magnitude only when a prompt token is actively attended to and exhibits above-average alignment with the optimization objective. This allows the formulation to highlight adversarial structural tokens (e.g., GCG suffixes) that carry benign value semantics but distort the attention routing distribution to bypass safety constraints.

Invoking the direct residual stream identity from Section 2, the final ranking metric used to isolate and construct the critical token subset $\mathcal{Q}$ is:

$$\text{Influence}(\mathbf{x}_i) = \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}} \right\|_2 \approx \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{x}_i} \right\|_2,$$

where $\frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(L-1)}}$ is computed via Equation (6).